Hamid Bostani

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Postdoctoral Researcher in Machine Learning Security at University of Luxembourg
Researcher in adversarial machine learning and systems security, focusing on trustworthy AI, adversarial robustness, and the security of AI-enabled systems.

CONTACT INFORMATION

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RESEARCH INTERESTS

- Adversarial Machine Learning
- Trustworthy AI
- AI-enabled Systems (e.g., Malware & Intrusion Detection Systems)

EDUCATION

Radboud University Nijmegen, The Netherlands. Oct. 2020 – Oct. 2024
Ph.D. in Computer Science.
Dissertation: **Rethinking the Security of Machine Learning in Malware Detection** (Supervisors: Prof. Veelasha Moonsamy and Dr. Erik Poll) - **Winner of 3rd VERSEN PhD Thesis Award 2025**
DOI: 10.54195/9789465151304
The dissertation was publicly defended on September 9, 2025. Watch the defense

Islamic Azad University (IAU), South Tehran Branch, Tehran, Iran. Sept. 2012 - Sept. 2015
M. Sc. in Computer Engineering (Software Engineering), GPA: 17.55/20.
Thesis: **Intrusion Detection and Identification of Attacks on the Internet of Things (IoT) Using a Combination of Machine Learning Methods** (Supervisor: Prof. Mansour Sheikhan) - **Winner of the Best IAU Master Thesis Award**
DOI: 10.13140/RG.2.2.28469.70884

Islamic Azad University (IAU), Shiraz Branch, Shiraz, Iran. Sept. 2004 - Sept. 2008
B. Sc. in Computer Engineering (Software Engineering), GPA: 16.73/20, Major GPA: 18.53/20.
Project: "**Developing E-Commerce Systems (a Case Study in a Mobile Phone Store)**" (Supervisor: Dr. Mostafa Fakhrahmad)

HONORS AND AWARDS

- **3rd Place, VERSEN PhD Thesis Award 2025 (**Dutch National Association for Software Engineering)
- **Best Employee Award** (NOET, Nov. 2019).
- **Research Funding** (Iran National Science Foundation, Feb. 2018).
- **Best Master's Thesis Award** (the 5th Research, Scientific & Technological National Festival of IAU, May 2017).
- **Outstanding Paper Award** (ICSPIS'2016).
- **Selected Paper** (IST'2016).

RESEARCH AND WORK EXPERIENCES

- **Postdoctoral Researcher** Aug. 2025–Present
  Security, Reasoning and Validation Research Group, Interdisciplinary Centre for Security, Reliability and Trust, University of Luxembourg, Luxembourg
- **Ph.D. Candidate** Oct. 2020–Oct. 2024
  Digital Security Group, Institute for Computing and Information Sciences, Radboud University, Nijmegen, The Netherlands.
- **Visiting Scholar** Oct. 2023–March 2024
  Cybersecurity Group, King's College University, London, UK.
- **Visiting Scholar** Oct. 2023–March 2024
  Systems Security Lab, University College London, London, UK.
- **Research Assistant** Aug. 2017–Sept. 2020
  Young Researchers and Elite Club of IAU, South Tehran Branch, Tehran, Iran.
- **Research Assistant** Aug. 2016–March 2017
  Research Center of Modeling and Optimization in Science and Engineering, IAU, South Tehran Branch, Tehran, Iran.
- **Full-Stack Developer** Sep. 2012– Sep. 2020
  Strategic Analysis and Information Security Group, Department of New Communications, National Organization for Educational Testing (NOET), Tehran, Iran.

- **Software Expert** April–Aug. 2012
  Software Systems Group, Department of Information and Communication Technology,
  Bank Hekmat Iranian, Tehran, Iran.

TEACHING ASSISTANT

- **AI and Cybersecurity**, University of Luxembourg (2025-2026)
- **Deep Learning**, Radboud University Nijmegen (2023)

PUBLICATIONS

Bibliographic indicators (Google Scholar October 2026) Citations: 826, h-index: 8.

Pre-prints

1. **H. Bostani** , M. Zainab , K. Tit, L. Cazzaro, and M. Cordy "Towards Certified Robustness of Feature-Engineered Malware Detectors via Randomized Smoothing", under peer review (2026).
2. M. Zainab, S. Dyrmishi, **H. Bostani**, L. Cavallaro, and M Cordy. "Breaking Windows Malware Detection: A Comprehensive Evaluation of Problem-Space Adversarial Robustness," under peer review (2026).
3. **H. Bostani**, J. Cortellazzi, D. Arp, F. Pierazzi, V. Moonsamy and L. Cavallaro. "On the Effectiveness of Adversarial Training on Malware Classifiers," under peer review (2026).

Journals

4. **H. Bostani** and V. Moonsamy. "Beyond Learning Algorithms: The Crucial Role of Data in Robust Malware Detection," IEEE Security & Privacy, vol. 23 (2025), no. 3, pp. 74-78. DOI: 10.1109/MSEC.2025.3550686 (Q1)
5. **H. Bostani**, Z. Zhao, Z. Liu, and V. Moonsamy. "Level Up with ML Vulnerability Identification: Leveraging Domain Constraints in Feature Space for Robust Android Malware Detection," *ACM Transactions to Privacy and Security*, vol. 28 (2025), no. 2, pp.1-32. DOI: 10.1145/3711899 (Q1)
6. **H. Bostani** and V. Moonsamy, "EvadeDroid: A practical evasion attack on machine learning for black-box android malware detection," *Computers & Security*, vol. 139 (2024). DOI: 10.1016/j.cose.2023.103676, Impact Factor (2024) = 5.6 (Q1)
7. **H. Bostani**, M. Sheikhan, B. Mahboobi, "A Strong Coreset Algorithm to Accelerate OPF as a Graph-based Machine Learning in Large-Scale Problems," *Information Sciences*, vol. 555 (2021), pp. 424-441. DOI: 10.1016/j.ins.2020.10.009, Impact Factor (2020) = 5.9 (Q1)
8. **H. Bostani**, M. Sheikhan, "Hybrid of Anomaly-Based and Specification-Based IDS for Internet of Things Using Unsupervised OPF based on Map-Reduce Approach," *Computer Communications*, Elsevier, vol. 98 (2017), pp. 52-71. DOI: 10.1016/j.comcom.2016.12.001, Impact Factor (2016) = 3.3 (Q1)
9. **H. Bostani**, M. Sheikhan, "Modifying Supervised Optimum-Path Forest in Intrusion Detection Systems Using Social Network Approaches and Unsupervised Learning," *Pattern Recognition*, Elsevier, vol. 62 (2017), pp. 56-72.DOI: 10.1016/j.patcog.2016.08.027, Impact Factor (2016) = 4.6 (Q1)
10. **H. Bostani**, M. Sheikhan, "Hybrid of Binary Gravitational Search Algorithm and Mutual Information for Feature Selection in Intrusion Detection Systems," *Soft Computing*, Springer, vol. 21 (2017), no. 9, pp. 2307-2324.DOI: 10.1007/s00500-015-1942-8, Impact Factor (2016) = 2.5 (Q2)
11. M. Sheikhan, **H. Bostani**, "A Security Mechanism for Detecting Intrusions in Internet of Things Using Selected Features Based on MI-BGSA," *International Journal of Information & Communication Technology Research*, Iran Telecommunication Research Center (ITRC), vol. 9 (2017), no. 2, pp. 53-62. http://journal.itrc.ac.ir/article-1-42-en.html

Conferences & Workshops

12. **H. Bostani**, Z. Zhao, and V. Moonsamy. "Improving Adversarial Robustness in Android Malware Detection by Reducing the Impact of Spurious Correlations," *In Proceedings of the 29th European Symposium on Research in Computer Security Workshops (**ESORICS 2024 Workshops**)*. DOI: 10.1007/978-3-031-82362-6_13
13. Z. Moti, A. Senol, **H. Bostani**, F.Z. Borgesius, V. Moonsamy, A. Mathur, G. Acar, "Targeted and Troublesome: Tracking and Advertising on Children's Websites," *In Proceedings of the 45th IEEE Symposium on Security and Privacy (**IEEE S&P 2024**)*. DOI: 10.1109/SP54263.2024.00118
14. **H. Bostani**, M. Sheikhan, B. Mahboobi, "Developing a Fast Supervised Optimum-path Forest Based on Coreset," *In Proceedings of 19th International Symposium on Artificial Intelligence and Signal Processing* (**AISP'2017**), pp. 172-177, 2017. DOI: 10.1109/AISP.2017.8324076
15. **H. Bostani**, M. Sheikhan, "Modification of Optimum-Path Forest using Markov Cluster Process Algorithm," *In Proceedings of 2nd of International Conference on Signal Processing and Intelligent Systems (**ICSPIS'2016**)*, pp. 1-5, 2016. (***Winner of the Outstanding Paper Award***) DOI: 10.1109/ICSPIS.2016.7869874
16. M. Sheikhan, **H. Bostani**, "A Hybrid Intrusion Detection Architecture for Internet of Things," *In Proceedings of 8th International Symposium on Telecommunication (**IST'2016**)*, pp. 601-606, 2016. (***Selected as one of the Best Papers***) DOI: 10.1109/ISTEL.2016.7881893

Book Chapters

17. H. Bostani, M. Cordy, and V. Moonsamy. "Machine Learning Security in Malware Detection: A Survey." Accepted for publication in *Intelligence and Security: Recent Developments*, Springer (2026).
18. M. Sheikhan and **H. Bostani**, "Hybrid and modified OPFs for intrusion detection systems and large-scale problems," in *Optimum-Path Forest* (pp. 109-136). Elsevier (2022). https://doi.org/10.1016/B978-0-12-822688-9.00013-X
19. M. Sheikhan, **H. Bostani**, "Binary Gravitational Search Algorithm (BGSA): Improved Efficiency," *in Encyclopedia of Information Assurance,* Taylor & Francis (2016). https://www.taylorfrancis.com/chapters/

ORAL/POSTER PRESENTATIONS

2024

- *Guest Talk:* **"Are We Truly Ready to Secure Malware Detection Against Adversarial Attacks?"**
  Machine Learning and Security Group, TU Berlin, Germany, December 2024.
- *Workshop Presentation:* **"Improving Adversarial Robustness in Android Malware Detection by Reducing the Impact of Spurious Correlations."**
  ESORICS 2024 Workshop on Security and Artificial Intelligence, Bydgoszcz, Poland, 20 September 2024. Watch the talk
- *Colloquium:* **"Rethinking Adversarial Machine Learning in the Context of Malware Detection,"**
  Lunch Colloquium, Digital Security Group, Radboud University, Nijmegen, The Netherlands, June 2024
- *Invited Talk:* **"Adversarial Robustness for Malware Classifiers."**
  KCL Cybersecurity Workshop, London, UK, February 2024.

2023

- *Webinar:* **"Adversarial Machine Learning: Challenges and Solutions in Malware Detection."**
  Iranian Society of Cryptology, December 2023. Watch the talk
- *Conference Presentation:* **"Improving Robustness of Machine Learning-based Android Malware Detection against Realizable Adversarial Examples."**
  ICT.OPEN'23, Utrecht, The Netherlands, April 2023.

2022

- *Colloquium:* **"Improving Robustness of Machine Learning-based Malware Detection against Realizable Android Adversarial Examples."**
  Lunch Colloquium, Digital Security Group, Radboud University, Nijmegen, The Netherlands, June 2022.

2020

- *Colloquium:* **"AI for Intrusion Detection."**
  Lunch Colloquium, Digital Security Group, Radboud University, Nijmegen, The Netherlands, October 2020.

2017

- *Conference Presentation:* **"Developing a Fast Supervised Optimum-path Forest Based on Coreset."**
  AISP'2017, Shiraz University, Shiraz, Iran, 25-27 October 2017.

2016

- *Conference Presentation:* **"Modification of Optimum-Path Forest using Markov Cluster Process Algorithm."**
  ICSPIS'2016, Amirkabir University of Technology, Tehran, Iran, 14-15 December 2016.
- *Conference Presentation:* **"A Hybrid Intrusion Detection Architecture for Internet of Things"**
  IST'2016, Iranian Research Institute for Information Science and Technology, Tehran, Iran, 27-28 September 2016.

ACADEMIC SERVICE

Program Committees

- 5th IEEE Conference on Secure and Trustworthy Machine Learning (SaTML) – 2026
- 1st Workshop on Real-world AI Security and Engineering for Cybersecurity Systems – 2026
- 8th ACM Workshop on Artificial Intelligence and Security (AISec) – 2025
- 7th ACM Workshop on Artificial Intelligence and Security (AISec) – 2024

Reviewer (Journals)

- ACM Transactions on Privacy and Security (TOPS) – 2025

Sub-Reviewer (Conferences)

- ICSE – 2025, 2026
- NDSS, IEEE Euro S&P – 2024
- USENIX Security – 2023

- USENIX Security, IEEE Euro S&P, ACM AsiaCCS – 2022
- CANS – 2021

TRAINING AND SKILLS

Training Courses

- **Education in a Nutshell**, Radboud University, Nijmegen, The Netherlands (2023).
- **Presentation Skills**, Radboud University, Nijmegen, The Netherlands (2023).
- **Summer School on Privacy-Preserving Machine Learning,** ITU Copenhagen and Aarhus University, Copenhagen, Denmark, (1-4 August 2022).
- **Advanced Conversation**, Radboud University, Nijmegen, The Netherlands (2021).
- **SQL Server Query Tuning and Optimization**, Faratar As Danesh Institute, Tehran, Iran (2018).
- **SQL Server 2016 – Design & Implementation**, Faratar As Danesh Institute, Tehran, Iran (2018).
- **Professional SCRUM Master**, Faratar As Danesh Institute, Tehran, Iran (2016).
- **MCSD Web Pack 2012**, Kahkeshane Noor Institute, Tehran, Iran (2016).
- **ETL (SSIS) and Data Mining (SSAS) 2012**, Faratar As Danesh Institute, Tehran, Iran (2014).
- **Data Warehousing & OLAP using SSAS 2012**, Faratar As Danesh Institute, Tehran, Iran (2014).
- **Win Application (C# & intro ADO.NET)**, South Industrial Management Institute, Shiraz, Iran (2008).

Computer Knowledge

- **Programming and Scripting:** Python, C/C++, C#, SQL, Java, JavaScript, HTLM & CSS, MATLAB.
- **Frameworks and Tools:** PyTorch, Scikit-Learn, Hugging Face Transformers, MATLAB Optimization, Fuzzy, and Neural Net Tools, Microsoft Visual Studio, SQL Server Management Studio, Visual Paradigm, Microsoft Office, Azure Boards.
- **Software Development Technologies:** C#.Net Windows Form, ASP.Net Web Form, APS.Net MVC, WCF Service, jQuery & AngularJS, ADO.NET Entity Framework, Java 2 Platform Micro Edition (J2ME), Microsoft BI Technologies (Data Quality, Integrated, Analysis, and Reporting Services).
- **Databases and Dataflow Systems:** SQL Server (programming).
- **Software Development Methodologies:** RUP, EUP, SCRUM.
- **OS:** Windows

Language Skills

- **Persian (Farsi):** Native language
- **English:** Fluent, TOEFL (Dec 2019) Internet-Based Test: 84/120 (Reading: 21/30, Listening: 22/30, Speaking: 21/30, Writing: 20/30)